

Qatar-1b

R-1813

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_150520 · created 2026-08-03T03:44:59+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457162.79823 +/- 0.00055 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.016
Transit depth (Rp/Rs)^2	2.97 +/- 0.56 [%]
Orbital Inclination (inc)	82.91 +/- 1.2
Airmass coefficient 1 (a1)	1.043 +/- 0.011
Airmass coefficient 2 (a2)	-0.0256 +/- 0.0063
Scatter in the residuals of the lightcurve fit is	1.09 %
Best Comparison Star	#1 - [395, 289]
Optimal Method	PSF photometry
Transit Duration (day)	0.0634 +/- 0.0089

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.172 ± 0.016 vs 0.1463 (deviation 1.61σ)
Duration fit vs published	0.0634 d vs 0.0692 d (deviation 0.65σ)
Mid-time O-C	-1.3 min
Depth SNR	10.7
Residual scatter	1.09 %

Light curve

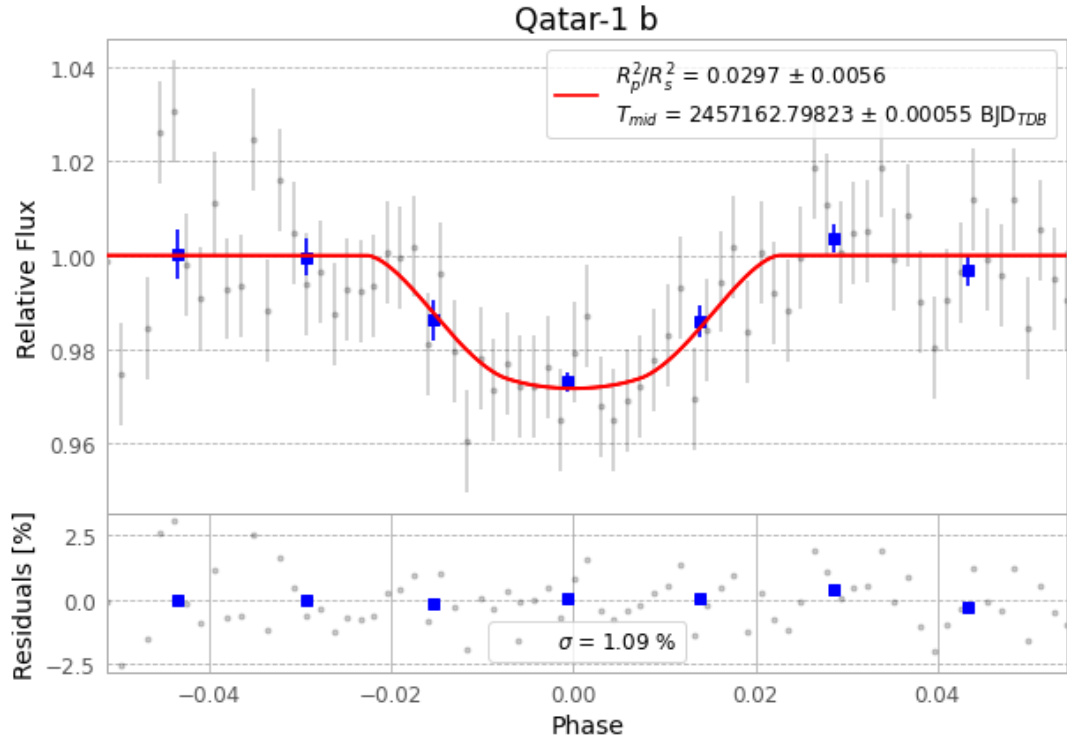


Figure 1 — FinalLightCurve_Qatar-1 b_2015-05-19.png (SHA-256 c8dff56b7f0d70b7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [281, 210], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2f27aa74c5e8465f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), Qatar-1150520052412.FITS → Qatar-1150520090012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-150520.log (9beee08a3f77...), FinalLightCurve_Qatar-1 b_2015-05-19.png (c8dff56b7f0d...), FinalLightCurve_Qatar-1 b_2015-05-19.csv (e40dd5d256b2...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 03:45Z.